

Stress transfer and dislocation response near a constrained $\text{Al}_{13}\text{Fe}_4$ inclusion in aluminium

I. Mikhailovskiy* D. Pshonkin

Moscow Polytechnic University, Moscow, Russia

Abstract

The mechanical response to a prescribed deformation of an $\text{Al}_{13}\text{Fe}_4$ inclusion in aluminium was examined using atomistic calculations. A ridge in a 91 428-atom interface cell was elongated by 0.194% and restrained while the matrix relaxed. The additional shear stress reached a magnitude of 15.45 MPa above the ridge axis, with substantial cancellation upon averaging across the cell. In matched loading runs, departure of an existing upper dislocation was detected at 113.6 MPa in the control and 122.7 MPa with the ridge deformed. Early motion and residual segments at the periodic seam limit attribution of this difference. A separate activated-glide model quantifies the sensitivity of creep to the assumed stress amplitude and activation volume. The calculations establish a conditional mechanical response; magnetic deformation of the inclusions and persistence after field removal remain untested.

Keywords molecular dynamics; interfacial stress; dislocations; aluminium alloys; constrained deformation

1 Introduction

Changes in the plastic response of aluminium alloys after magnetic exposure motivate the examination of mechanical interactions between inclusions and dislocations. Experiments on heterogeneous Al–Mg–Si alloys containing iron-bearing inclusions reported changes in creep after exposure to a static field [1–3]. In these experiments the treatment and mechanical loading were separated in time. Subsequent work also reported changes in transient creep and stress relaxation [4]. Any explanation of the retained response therefore requires a material state that persists after the treatment has ended.

A possible contribution is the stress generated when an inclusion changes shape within the surrounding matrix. Magnetostriction could provide the deformation if the inclusion contains an appropriate magnetic constituent. Its mechanical consequences depend on particle shape, elastic accommodation and crystallographic orientation. A scalar estimate of interfacial stress does not determine the shear acting on a particular dislocation. The pre-existing interface field and the distribution of obstacles also affect the response of a line placed near the inclusion.

The present calculations address the mechanical part of this problem using an aluminium matrix and an $\text{Al}_{13}\text{Fe}_4$ inclusion. A prescribed elongation of 0.194% sets the amplitude of a controlled perturbation. This value follows the modelling estimate associated with the earlier 147 MPa stress scale [3]; it has not been measured for the inclusions under magnetic exposure. The applied constraint fixes reference coordinates for the inclusion atoms. Consequently, the resulting field is the response to a specified mechanical boundary condition. Identification of that condition with a magnetic transformation requires additional evidence.

The study combines a paired static stress calculation with trajectories of existing dislocations in the same interface geometry. Attention is given to the distinction between signed shear near

*Corresponding author. ilyamihailovsy@gmail.com

the inclusion axis and averages across the simulation cell. The trajectories are examined through their initial relaxation, subsequent motion and interaction with the periodic seam. A separate activated-glide estimate then relates assumed stress amplitudes to a conditional change in creep rate. This restricted scope allows the calculated response and its limitations to be stated without assigning a microscopic origin to the experimental memory effect.

2 Methods

2.1 Geometry and imposed deformation

Calculations used LAMMPS, release 22 July 2025, with KOKKOS/CUDA acceleration [5] and the Al–Si–Mg–Cu–Fe second-neighbour modified embedded-atom potential of Jelinek et al. [6]. The face-centred cubic aluminium matrix has lattice parameter $a_0 = 4.05 \text{ \AA}$ and axes $x \parallel [1\bar{1}0]$, $y \parallel [11\bar{2}]$ and $z \parallel [111]$. This orientation places a (111) glide plane parallel to the nominal interface and its $[1\bar{1}0]$ glide direction along x . An edge line parallel to y then carries Burgers vector $\mathbf{b} = (a_0/2)[1\bar{1}0]$, with magnitude $b = 2.864 \text{ \AA}$.

The monoclinic $\text{Al}_{13}\text{Fe}_4$ structure [7] forms a $20 \text{ \AA}$ support layer topped by a half-elliptical ridge with semi-axes $35 \text{ \AA}$ along x and $20 \text{ \AA}$ along z . The ridge spans the periodic y direction. The interface cells contain 91 428 atoms, comprising 72 532 matrix atoms and 18 896 inclusion atoms. Their in-plane dimensions are $154.64 \times 64.48 \text{ \AA}^2$. Periodic boundaries apply along x and y ; the upper aluminium surface faces vacuum and the bottom $6 \text{ \AA}$ is fixed. Figure 1 shows the geometry. Interfacial bonding and any atomic rearrangement across the interface are governed by the interatomic potential.

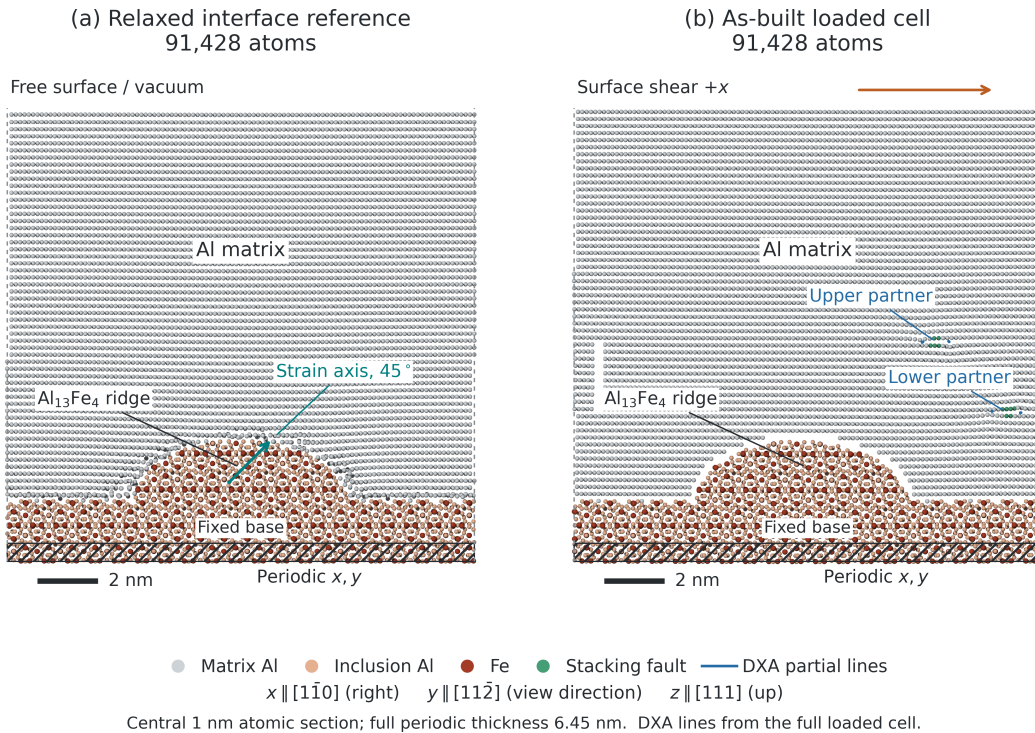


Figure 1: Atomistic interface geometry rendered from LAMMPS configurations with OVITO. The $\text{Al}_{13}\text{Fe}_4$ ridge extends along y above a flat support layer. The coordinate axes identify the glide direction, line direction and interface normal. The loaded configuration includes an oppositely signed edge-dislocation pair. The upper layers transmit the applied shear and the bottom layer is fixed.

The ridge atoms are displaced about their centroid using the small-strain tensor

$$\mathbf{E}_0 = \frac{3\varepsilon_0}{2} \left(\mathbf{u} \otimes \mathbf{u} - \frac{\mathbf{I}}{3} \right), \quad \varepsilon_0 = 0.00194, \quad \mathbf{u} = \frac{(1, 0, 1)}{\sqrt{2}}. \quad (1)$$

Its zero trace conserves volume to first order. The chosen direction maximises the imposed xz shear component, $E_{0,xz} = 0.001455$, within this family of deformations. It defines one favourable orientation. The support layer receives no imposed displacement.

After preparing a relaxed reference structure, control and perturbed copies are minimised with identical settings. Inclusion atoms are attached to their respective reference coordinates by harmonic springs of stiffness $20 \text{ eV}/\text{\AA}^2$. The matrix remains free to relax. The potential itself is unchanged, so the constraint supplies forces needed to maintain the deformation. A material eigenstrain would instead alter the stress-free state of the inclusion. The spring reactions remain part of the paired response even though the same stiffness is used in both calculations. Full preparation and convergence information is given in the Supplementary Information.

Preparation includes conjugate-gradient minimisation, 6 ps at 300 K, 12 ps of cooling to 0.5 K and a final minimisation. The static pair is then constructed from one prepared control, with matching atom identities. Both copies reach the requested force two-norm of $0.02 \text{ eV}/\text{\AA}$ after their separate minimisations. This pairing reduces differences arising from independently relaxed interfaces. It does not demonstrate insensitivity to the initial structure or to minimisation tolerance; the reported comparison is for this one prepared state.

2.2 Stress and dislocation measurements

The six components of the configurational atomic virial are averaged before forming stress invariants. The conversion uses the aluminium atomic volume $16.61 \text{ \AA}^3$. Profiles compare the perturbed and control tensors, $\Delta\sigma = \sigma_\varepsilon - \sigma_0$, in layers $4 \text{ \AA}$ thick. The stored tensor uses positive normal stress in compression. Its sign is opposite to the tensile-positive Cauchy convention; absolute shear projections and invariants are unchanged. The coordinate $r = z - 20 \text{ \AA}$ measures height above the nominal interface plane; $d = r - 20 \text{ \AA}$ is height above the nominal ridge crest. For a layer spanning the cell, d is not a nearest-surface distance.

Two spatial averages are retained, across the full cell width and within $|x - x_c| < 10 \text{ \AA}$ of the ridge axis. Layers intersecting the inclusion are excluded from the matrix profile. The signed shear on the selected $(111)[\bar{1}\bar{1}0]$ system is $\Delta\sigma_{xz}$. The supplementary scalar $\text{RSS}_{\max}$ is the maximum of $|\mathbf{s} \cdot \Delta\sigma \mathbf{n}|$ over the twelve fcc slip systems, with unit slip direction $\mathbf{s}$ and plane normal $\mathbf{n}$.

Layer membership and the axial window are defined by control coordinates and applied to the same atom identities in both snapshots. This construction avoids changing the sampled population as atoms move across layer boundaries. The retained layers lie entirely above every inclusion atom in both snapshots and contain only matrix aluminium, identified by atom ID. No thermal kinetic term enters the minimised profiles. The von Mises stress of the difference tensor and the difference between two von Mises stresses are distinct quantities; only the former describes the shear invariant of the perturbation.

The loaded interface cells also contain 91 428 atoms. Two oppositely signed edge lines are inserted on parallel glide planes separated by $23.4 \text{ \AA}$, with an initial horizontal offset of the same magnitude. Their motion is followed using the dislocation extraction algorithm (DXA) [8] in OVITO [9]. Dynamics uses a 1 fs step and a 300 K Nosé–Hoover thermostat. After 5 ps without applied shear, forces on the upper atomic layers increase according to a ramp whose rate is smoothly introduced over 16 ps. The matched restrained runs end at nominally 145.45 MPa. An additional unstrained run with an unrestrained inclusion extends to 400 MPa. Analysis samples are spaced by 2 ps, corresponding to approximately 9.09 MPa in the linear part of the ramp.

Line positions are unwrapped along the periodic x direction and grouped by Burgers-vector family. For an individual line, a linear baseline is fitted to available positions from 4 ps onward

while the nominal stress is below 40 MPa. Departure requires a deviation larger than six times the baseline residual standard deviation, with a minimum of 8 Å, and persistence in the next two available sampling positions. The stored detector also accepts a missing line at a following frame. Accordingly, line continuity is examined alongside this numerical criterion, especially when a segment reaches the periodic seam. Quoted intervals represent sampling brackets rather than confidence intervals.

A separate 92 800-atom alloy cell provides a contextual mobility comparison. It contains 1.0 at.% Mg and 0.6 at.% Si and no inclusion. Its distinct geometry and stepwise loading are documented in the Supplementary Information. This comparator is not included in the matched interface calculation.

3 Results

3.1 Stress transferred to the matrix

The unstrained interface already supports a substantial residual stress. In the nearest retained layer, at $d = 2$ Å, the control von Mises stress is approximately 850 MPa. The next three layers give 106, 185 and 157 MPa. Such spatially averaged atomic stresses reflect the assembled interface and its relaxed atomic structure. Their magnitudes cannot be equated directly with a macroscopic uniaxial yield stress. The non-monotonic near-interface profile also prevents extraction of a unique decay length from the absolute stress alone.

The imposed perturbation produces a smaller change superposed on that background (Fig. 2). The signed axial-window shear is negative throughout the retained profile, reaching -15.451 MPa at $d = 22$ Å. Its magnitude is about 11–15 MPa over the sampled interval $d = 2$ –30 Å, then falls to 10.258 MPa at 46 Å, 5.139 MPa at 66 Å and 1.905 MPa at 86 Å. These values describe spatial attenuation in a minimised configuration. They contain no measurement of a relaxation time.

Across the full width, the signed shear is mostly positive and reaches 5.042 MPa at $r = 50$ Å. Its different sign and smaller magnitude show that the averaging region changes the mechanical quantity being reported. At $r = 46$ Å, the width-averaged xz component nearly cancels, while $\text{RSS}_{\text{max}} = 0.558$ MPa occurs on $(111)[01\bar{1}]$. The nearest layer, at $r = 22$ Å, instead selects $(\bar{1}\bar{1}\bar{1})[101]$ and gives the largest width-averaged RSS_{max} , 5.651 MPa. The remaining layers select $(111)[\bar{1}\bar{1}0]$. The local minimum is therefore partly a consequence of tensor projection and spatial averaging.

In the thirteen layers at $r > 60$ Å, the width-averaged RSS_{max} has mean 2.477 MPa and standard deviation 0.543 MPa. These are spatial statistics of this configuration. They provide no independent estimate of numerical noise or a detection limit. The small rise in the width average at larger distances accompanies a continued decrease of the axial-window magnitude. A finite slab and its free surface can influence this profile, and the available calculation does not separate their contributions. No vanishing-field distance is assigned.

An affine fit to the iron sublattice above the support gives a projection of 0.9728 onto the imposed ridge deformation, with formal fit standard error 0.0038. The springs thus maintain most of the selected deformation through matrix relaxation. The fit error describes residuals within one configuration and cannot represent uncertainty across independent simulations. A continuum comparison using a spherical inclusion with the same nominal tensor and identical isotropic elastic constants inside and outside gives an interior maximum resolved shear of 41.45 MPa [10]. This separate boundary-value problem provides a reference stress scale. It supplies no upper bound for the constrained atomistic ridge, whose support layer, free surface and spring reactions are absent from the sphere model.

Prescribed strain $\varepsilon = 0.194\%$; $\Delta\sigma = \sigma_{\text{strained}} - \sigma_{\text{control}}$

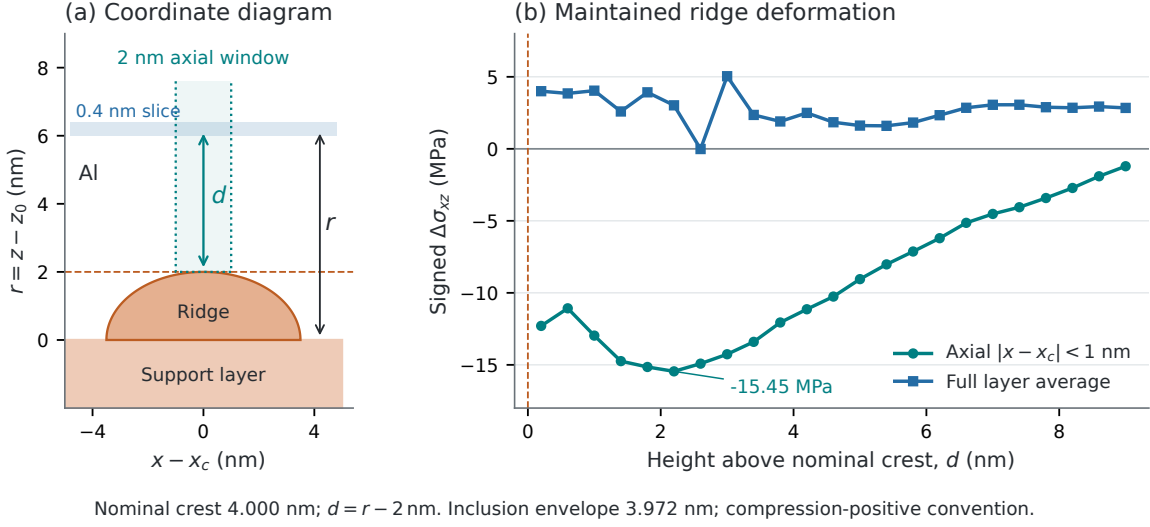


Figure 2: Distance convention and signed shear in the paired minimised configurations. The coordinate diagram defines r from the flat interface and d from the nominal ridge crest at $z = 40$ Å. The stress panel compares $\Delta\sigma_{xz}$ in the 20 Å axial window with its average across the cell width. Only layers whose lower edge lies above all inclusion atoms in both snapshots ($z_{\text{max}} = 39.719$ Å) are retained. The compression-positive convention is used. The opposite signs show the sensitivity of the reported stress to the averaging region.

3.2 Existing dislocations during loading

The pre-existing lines change position during the start of all three ramps. Figure 3 shows the matched restrained upper-line histories. In the unrestrained control, the upper line moves approximately 38 Å towards the ridge axis during the first 6 ps. In the restrained control and perturbed cells, the corresponding displacements are 31.0 and 33.2 Å. The upper line then lies within about 7–10 Å of the axis. Loading starts at 5 ps, so this initial motion occurs predominantly during preconditioning and the smooth onset of the load.

These displacements demonstrate that the inserted pair is not stationary in its initial environment. Residual interface stresses, interaction between the lines, periodic images and boundary forces are all present. Because both restrained configurations move substantially at the outset, their later response must be interpreted relative to the relaxed line positions. The early interval does not isolate motion caused solely by the imposed ridge deformation.

For the upper line, departure from its low-stress trend is detected at 113.6 MPa in the restrained control and 122.7 MPa in the perturbed cell. The preceding sampled stresses are 104.5 and 113.6 MPa, respectively. Thus the criterion is crossed in adjacent sampling intervals. Each configuration has only one trajectory at one loading rate. This observation leaves the size and reproducibility of a deformation-induced shift unresolved. A statistical null effect or a material critical stress cannot be extracted from it.

The lower line in both restrained runs is no longer separately identified by 10 ps. Later DXA segments appear near the periodic seam. Assigning these segments to the original line by Burgers-vector family alone would create a spurious late departure event. At the final analysed time of 44 ps, about 140.9 MPa, the perturbed cell has no extracted segment. The control retains three segments with combined length 39.3 Å near $x = 7$ –10 Å. Their presence precludes describing both configurations as dislocation-free. Short segments are reported as local DXA signals without identifying a nucleation mechanism.

In the unrestrained control, the lower line crosses the motion criterion between 95.5 and

91,428 atoms; inclusion held in both runs; 300 K

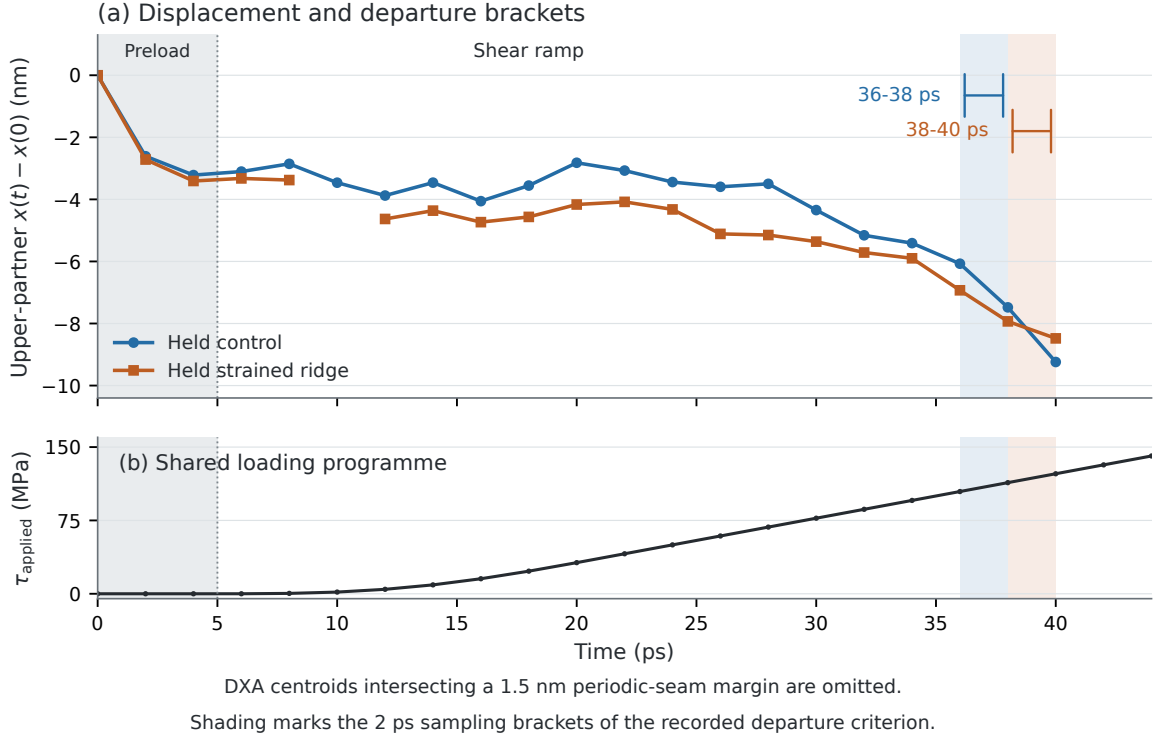


Figure 3: Existing-line motion in the interface cells. Upper-line positions are followed through preconditioning and shear loading, with the applied stress shown alongside. The restrained control and perturbed cases cross the departure criterion one saved frame apart. The initial motion towards the ridge and the retained DXA segments at the periodic seam constrain interpretation of the later trajectories. Segment disappearance or reassignment is not evidence of a new interface dislocation.

104.5 MPa. The analysed trajectory extends to 395.45 MPa at 100 ps. No newly formed, persistent interface dislocation is confirmed within those sampled configurations. Short signals also occur at the periodic seam at higher loads. The nominal run endpoint is 400 MPa, beyond the last analysed regular frame. This single run gives a restricted observation of interface activity under its own boundary conditions; it does not measure a universal nucleation threshold.

4 Conditional connection to creep

A scalar activated-glide model is used to examine the consequences of the computed stress scale. It assumes dilute compact particles with exterior shear amplitude $\tau(R) = \tau_m(a/R)^3$, where R is measured from a particle centre. Positive and negative local stresses are assigned equal weights, and their effect on an elementary activated rate is represented by $\cosh(V^*\tau/k_B T)$ [11]. Here V^* describes the stress sensitivity of the activation barrier. These assumptions give the fractional rate increase

$$H = fx \int_0^x \frac{\cosh u - 1}{u^2} du, \quad x = \frac{V^* \tau_m}{k_B T}, \quad (2)$$

to leading order in inclusion volume fraction f . The tensor field, grain orientations and interactions between particles are reduced to a single assumed amplitude.

The evaluation uses $T = 300$ K and $f = 0.00246$ as a representative dilute scenario. A target $H = 0.25$ serves as a reference motivated by the scale of the reported creep changes. Activation

volumes from $19b^3$ to $142b^3$ are explored as a sensitivity range. The value $70b^3$, reported for a differently processed Al–Mg–Si alloy [12], provides context for that range. No activation volume is determined by the present trajectories.

The chosen fraction corresponds approximately to a model mass fraction of 0.35% with assumed inclusion and matrix densities of 3.85 and 2.70 g/cm³, respectively. These inputs describe a representative dispersion and carry their own compositional uncertainty. Moreover, H is a rate change in the activated model. Relating it to accumulated experimental creep also requires a loading history over which that rate description applies. Treating the experimental change as a fixed target therefore introduces a second level of modelling beyond the atomistic stress calculation.

Solving Eq. (2) gives required amplitudes from 62.69 to 8.39 MPa as V^* increases across the selected range (Fig. 4). If 15.451 MPa is inserted as a representative amplitude, $H = 0.25$ occurs at $V^* = 77.09b^3$. Using 41.45 MPa instead gives $28.74b^3$. The conditional range for these inputs is therefore about 29–77 b^3 . At $70b^3$, the 15.451 MPa scenario gives $H = 0.156$.

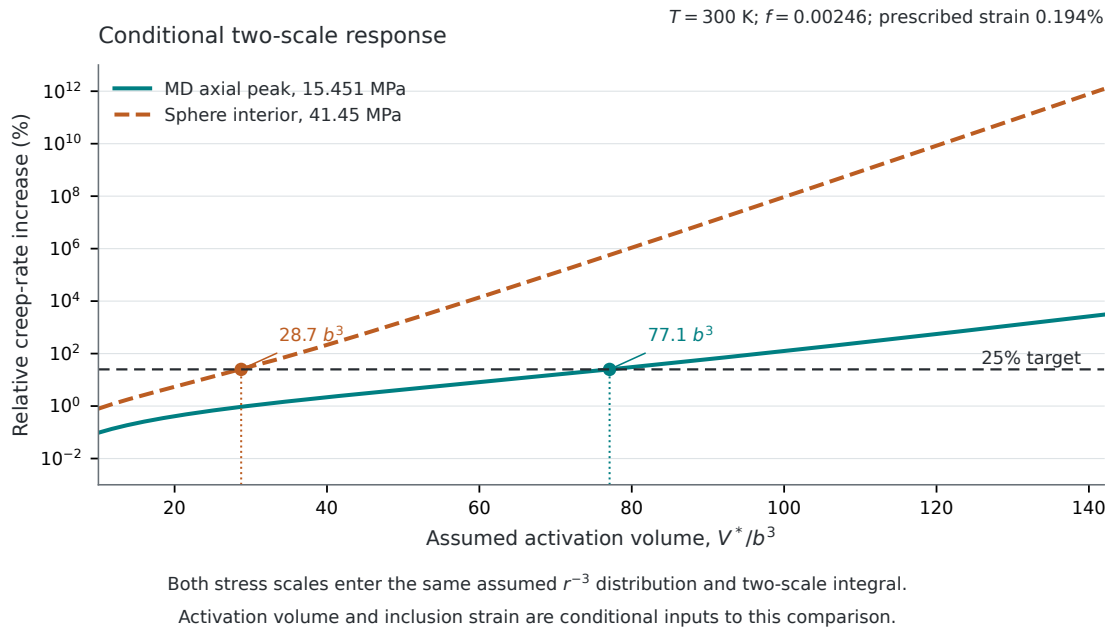


Figure 4: Conditional activated-glide estimate at 300 K and $f = 0.00246$. The required shear amplitude for a 25% rate increase depends strongly on activation volume. The amplitudes 15.451 and 41.45 MPa are taken as separate illustrative inputs from the ridge profile and sphere interior. Their use in the scalar spatial model is an additional assumption. The curve is not fitted to a measured activation volume or a simulated creep rate.

The axial ridge maximum and the sphere interior value belong to different physical and spatial definitions. Substituting either into Eq. (2) tests the consequences of an assumed amplitude; it does not establish the exterior surface stress of real particles. The exponential dependence on $V^*\tau_m$ permits the target change to be obtained by many combinations of uncertain inputs. Agreement at a selected activation volume therefore supplies no independent validation of a magnetic mechanism.

5 Discussion

The paired static calculation quantifies how a specific constrained deformation changes the matrix stress. Retaining its sign reveals opposing contributions that are obscured in a full-width scalar average. This distinction matters when assessing force on a dislocation whose position selects

only part of the field. At the height reached by the upper line after preconditioning, the axial-window shear change is approximately 13–14 MPa. Its scale is comparable to the separation between sampled departure stresses. A dedicated series of trajectories would be required to distinguish systematic changes from sensitivity to initial state and line identification.

The current model includes several restrictions on that interpretation. The ridge is extended periodically along the line direction and rests on a constrained support. It does not represent an isolated particle in a macroscopic polycrystal. The interaction potential has broad elemental coverage, but transferability to this interface, its defect cores and solute interactions has not been independently established here. No cell-size study, loading-rate series or ensemble of velocity seeds supports extrapolation of the recorded departure stresses. The thermostat also acts during loading, so these runs do not determine adiabatic heating or the heat-release changes measured experimentally.

The nanometre span of the present profile is tied to the imposed ridge dimensions and to the finite simulation domain. Elastic inclusion solutions organise distance through the ratio of separation to particle size. A larger geometrically similar particle can therefore act over a larger absolute distance at the same dimensionless strain. Applying that scaling to this particular constrained ridge requires further checks of geometry and boundary effects. The calculated profile alone cannot establish stress overlap between inclusions several micrometres apart.

The alloy comparator adds evidence of limited line mobility in one random solute configuration. The tracked line accumulates 1.98 Å net displacement during the completed loading history through 75 MPa, below one Burgers-vector magnitude. Its configuration, time window and internal stresses differ from those of the interface runs. It therefore provides no transferable pinning constant or calibration of V^* . The trajectory, analysis procedure and limitations of the comparator are described in the Supplementary Information.

Connecting the imposed deformation to magnetism remains a separate question. Studies of $\text{Al}_{13}\text{Fe}_4$ report weak or unusual paramagnetic behaviour rather than establishing the strongly ferromagnetic response assumed for some iron-bearing inclusions [13, 14]. The phase composition, magnetic response and field-induced shape change of the actual inclusions need to be established together. The chosen interatomic potential and coordinate restraints supply no magnetic constitutive relation. Consequently, neither the prescribed strain nor the calculated stress is a measurement of magnetostriction.

The experimental sequence also places a temporal condition on any mechanism. Creep is measured after magnetic exposure, whereas the static calculation maintains the mechanical constraint and the dynamic runs do not reproduce its removal followed by delayed loading. Irreversible defect rearrangement, retained deformation and changes in obstacle populations could each alter the subsequent response, but none is established by these records. The results identify mechanical quantities that such an explanation would need to reproduce. They leave the origin and lifetime of the experimental memory open.

6 Conclusions

A constrained 0.194% deformation of an $\text{Al}_{13}\text{Fe}_4$ ridge produces a signed axial shear change with peak magnitude 15.45 MPa in the 91 428-atom interface cell. Spatial cancellation lowers the full-width response and changes its sign. The sampled axial magnitude decreases to about 2 MPa at a height of 8.6 nm above the crest.

The matched loading trajectories show substantial initial motion of existing lines and an upper-line departure difference of one saved frame. Periodic-seam segments and limited sampling restrict quantitative attribution. The separate activated-glide calculation expresses sensitivity to assumed stress and activation volume. Together these results characterise the response to a maintained mechanical constraint and delimit its interpretation as a possible contribution to magnetically modified creep.

Author contributions

I. Mikhailovskiy contributed methodology, software, investigation, formal analysis, visualisation and the original draft. D. Pshonkin contributed conceptualisation, resources, supervision, and review and editing.

Data availability

The repository <https://github.com/iluuua/magnetostriction-plasticity-bounds> provides the paired static configurations with atomic virials, the position trajectories of the three interface loading runs, their initial and final configurations, and simulation logs. The publication manifest specifies the retained trajectory columns and sampling. The alloy comparator includes its complete saved atomistic coordinate and stress history, the derived 131-frame position table, summaries, initial and final structures, and log. The Supplementary Information maps the reported quantities to the archived records.

Code availability

The LAMMPS inputs, launcher for the deposited initial configurations, analysis and figure-generation code are available in the same repository. The publication manifest identifies the files used for the reported calculations.

Declaration of competing interest

The authors declare no competing financial interests.

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